

Roshan Kumar Gupta

Research Engineer & Systems Architect

CAREER OBJECTIVE

Motivated Computer Science Engineering student with a strong foundation in low-level systems architecture, high-performance computing, and deterministic runtimes. Deeply passionate about building robust, physics-aware solutions and scalable platforms that solve complex engineering challenges. Seeking a Research Engineer or Systems Architect Intern role where I can leverage my proficiency in C++, Python, and parallel computing architectures to drive innovation and push the boundaries of modern computational frameworks.

EDUCATION

- Bachelor of Engineering (Computer Science)**
Chandigarh University, Mohali | CGPA: 6.98 (2025 – 2029)
- Senior Secondary Education (CBSE)**
RSVM | Percentage: 65% (2023 – 2025)
- Secondary Education (CBSE)**
GGPS | Percentage: 80.4% (2022 – 2023)

WORK EXPERIENCE

Earthos Platform | Founder & Lead Architect (Jul 2026 – Present)

- Deployed the production storefront for *The AGI Question* using Next.js 16 and Supabase.
- Engineered automated multi-warehouse nearest-node shipping algorithms via Shiprocket APIs.
- Developed a secure PDF generation pipeline with real-time DRM watermark fingerprinting.

SMG Electric Scooters Ltd | L3 IT Intern (Dec 2025 – Jan 2026)

- Designed an internal CRM portal, streamlining logistics and vendor workflows via automation.

Chandigarh University | Student Rep (Oct 2025 – Present)

- Managed high-stakes communication to resolve academic and infrastructure issues for 400+ students.

PROJECTS

Project TARS Ecosystem | Python, Data Science (2025) tars.earth

- Secured \$5,500 Emergent Ventures grant; designed the TARS Core physics-constrained validation framework achieving 70.1% precision on NASA TESS data.
- Engineered *tars_infra*, a parallel FITS data ingestion engine utilizing multi-drive spanning to index 150,000+ files from the MAST database.

FieldChain | C++, Vulkan (2025) github.com/rk-roshan-kr/FieldChain

- Secured \$1,500 grant; engineered a GPU-centric storage layer sustaining 7GB/s sequential writes via PCIe 4.0, outperforming specialized DPUs by 4.2x via Vulkan.

Project Earthos (Synapse Arch) | Python, AI (2026) earthos.shop

- Architected a neural-symbolic cognitive substrate combining event-sourced memory hydration.
- Implemented AST locks to block non-deterministic floating-point namespaces, enforcing execution parity.

ACHIEVEMENTS

- Grants (2025):** Secured cumulative research grants totaling \$6,500 (Emergent Ventures, FieldChain).
- Conferences (2026):** Awarded Best Student Paper for FieldChain research at IEEE CCNCPS 2026 in Dubai.

PERSONAL INFO

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TECHNICAL SKILLS

- Low-Level -**
CUDA, Vulkan, PTX, C++, Deterministic Runtimes
- Systems & ML -**
React, PyTorch, Next.js
- Backend -**
Node.js, PostgreSQL, Supabase, REST APIs

INTERPERSONAL

- Adaptable in fast-paced environments
- Active Listening & Clear Communication
- Leadership & Strategic Project Planning
- Problem Solving with a Critical Mindset

AWARDS & CERTS

- Research Excellence Award** (CU)
- Winner**, ZeroToOne Hackathon
- Winner**, Zinnovatio Hackathon
- Scientific Writing (Polytechnique)
- Azure AI Cert (Vision, NLP)

EXTRACURRICULAR

- Hackathons -** Highly active in the global hackathon circuit (15+ events).
- Open Source -** Contributor to physics simulation libraries.
- Research -** Active member in deep tech systems architecture groups.

INTEREST & HOBBIES

- Competitive Programming
- Quantum noise limits
- Astronomical data
- Hardware tinkering
- Reading research

LANGUAGES KNOWN

- English - Professional Working
- Hindi - Native or Bilingual